

KARL POPPER · 1902-1994

CLARENDON PRESS · 1972

AN EVOLUTIONARY APPROACH

Objective Knowledge

Nine essays arguing that knowledge is not a state of mind, that induction does not exist, and that we grow by killing our own guesses.

Popper opens chapter one with a sentence almost no philosopher would risk: *I think I have solved a major philosophical problem: the problem of induction.* The solution is not a better justification for generalising from experience. It is the claim that **we never generalise from experience at all — we guess, and then we try to destroy the guess.** Everything else in the book follows from taking that seriously: a theory of mind, a theory of evolution, a theory of what science is for, and an account of knowledge as something that exists outside anybody's head.

CHAPTERS

9 + appendix

PAPERS, 1965-71

CORE MOVE

Conjecture

THEN REFUTATION

NEW HERE

World 3

KNOWLEDGE WITHOUT A KNOWER

DOWNSTREAM

Deutsch

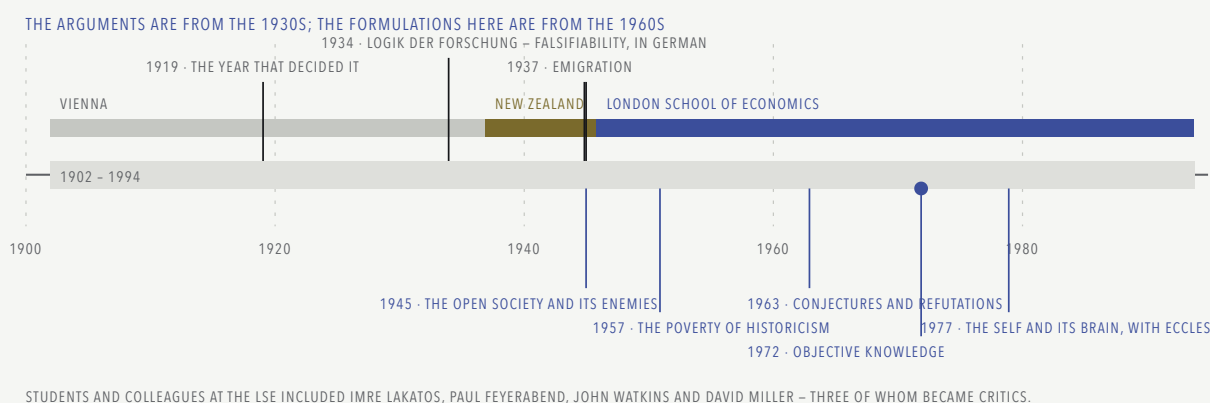
ALMOST ENTIRELY

Objective Knowledge is a late book — Popper was seventy when it appeared, and the papers it collects were written between 1965 and 1971. Nearly everything in it is a development of positions he had held since his twenties, and those positions were formed in a specific place at a specific moment: Vienna between the wars, where two things were simultaneously true. Einstein had just overturned Newton, and half the intellectual class had adopted doctrines that seemed to explain everything and could be refuted by nothing.

Popper was born in 1902 to a Viennese family of Jewish descent that had converted to Lutheranism — a background that made him, in the 1930s, exactly the wrong sort of person to remain in Austria. His father was a lawyer with a fourteen-thousand-volume library. Popper left school early, spent time as a manual worker, and apprenticed himself to a cabinetmaker, an experience he described as more instructive than most of his formal education; the master craftsman he worked under, Adalbert Pösch, appears in his autobiography as a man who claimed to know everything and could not be told anything. He qualified as a schoolteacher, took a doctorate in the psychology of thought in 1928, and taught in Viennese secondary schools while writing the book that made his name.

The formative year was 1919, when he was seventeen. He had been briefly attracted to Marxism, and broke with it after an incident in which unarmed young socialists — some of them his friends — were shot by police during a demonstration the party had encouraged. What appalled him was not only the deaths but the ease with which the theory absorbed them: the episode was immediately explained as a necessary stage, confirming the doctrine rather than testing it. In the same year, Eddington's eclipse expedition tested Einstein's prediction that starlight would bend by a specific amount near the sun. **Popper noticed that the two cases had opposite structures, and spent the rest of his life on the difference.**

PLATE 01 NINETY-TWO YEARS, AND WHERE THIS BOOK SITS



The gap between 1934 and 1959 is a translation delay: *Logik der Forschung* appeared in English as *The Logic of Scientific Discovery* only twenty-five years after publication, which is why anglophone philosophy encountered Popper's ideas out of order and often through his critics.

THE OFFICIAL OPPOSITION

Popper is routinely described as a member of the Vienna Circle, and he was not. The Circle — Moritz Schlick, Rudolf Carnap, Otto Neurath and others — were logical positivists: they held that a statement is meaningful only if it can be verified by experience, which made metaphysics not false but literally

nonsense. Popper knew them, published in their book series, and disagreed with them about nearly everything of consequence. Neurath called him *die offizielle Opposition*, and the label stuck because it was accurate.

The disagreement matters for reading this book, because Popper's targets are often positivist rather than commonsensical. Where the positivists sought a criterion of *meaning*, Popper sought a criterion of *scientific status* — and insisted that unscientific does not mean meaningless. Metaphysical ideas, on his account, are frequently the seedbed of scientific ones: atomism was metaphysics for two thousand years before it was physics. Where the positivists wanted to ground science in observation, Popper argued that observation is theory-laden and cannot ground anything. And where they hoped an inductive logic could assign probabilities to theories, he thought the whole programme was chasing something that does not exist.

PLATE 02 POPPER AGAINST THE CIRCLE, POINT BY POINT

THE VIENNA CIRCLE		POPPER
VERIFIABILITY	THE CRITERION	FALSIFIABILITY
MEANINGFUL FROM MEANINGLESS	SORTS	SCIENTIFIC FROM NON-SCIENTIFIC
PROTOCOL SENTENCES – BEDROCK OBSERVATION	THE BASIS	PILES DRIVEN INTO A SWAMP; NO BEDROCK EXISTS
NONSENSE, TO BE ELIMINATED	METAPHYSICS	OFTEN MEANINGFUL, AND THE PARENT OF SCIENCE
A LOGIC OF CONFIRMATION IS ACHIEVABLE	INDUCTION	THERE IS NO SUCH LOGIC AND NONE IS NEEDED

THE SECOND ROW IS THE CRUCIAL ONE. POPPER IS NOT SAYING UNSCIENTIFIC CLAIMS ARE WORTHLESS – HE IS SAYING THEY ARE A DIFFERENT KIND OF THING.

Popper later claimed to have "killed logical positivism," and while that overstates a single person's role, the verifiability criterion did collapse – partly because it could not survive its own test, being neither verifiable nor analytic. The demarcation problem he substituted for it is still live.

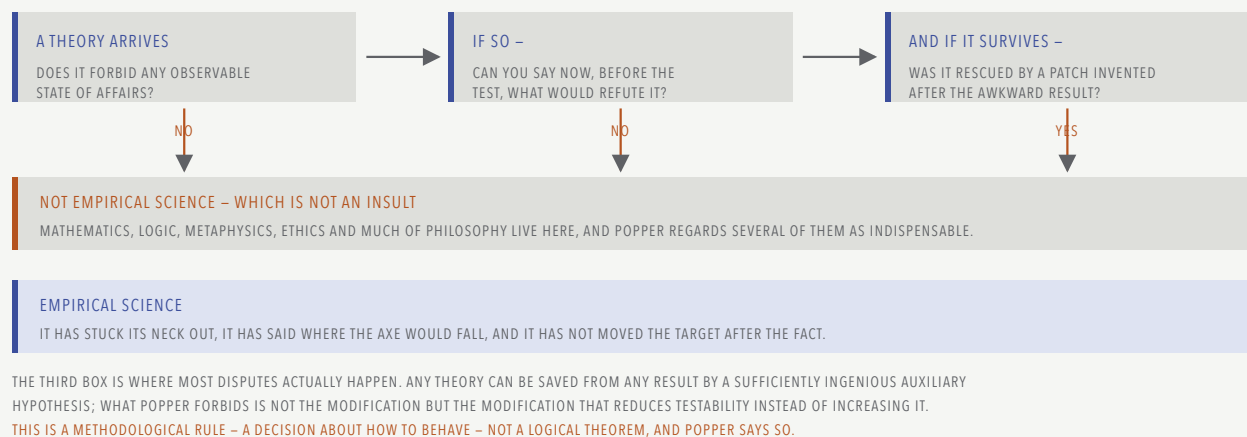
What Makes a Theory Scientific *Not that it explains a lot. That it forbids a lot.*

This is the idea Popper is best known for and the one most often misstated, including by people who invoke it approvingly. It is not a theory of meaning, not a theory of truth, and not a claim that unfalsifiable statements are worthless. It is a proposal for where to draw one particular line: **a theory belongs to empirical science if there is some conceivable observation that would count decisively against it, and its author can say in advance what that observation is.**

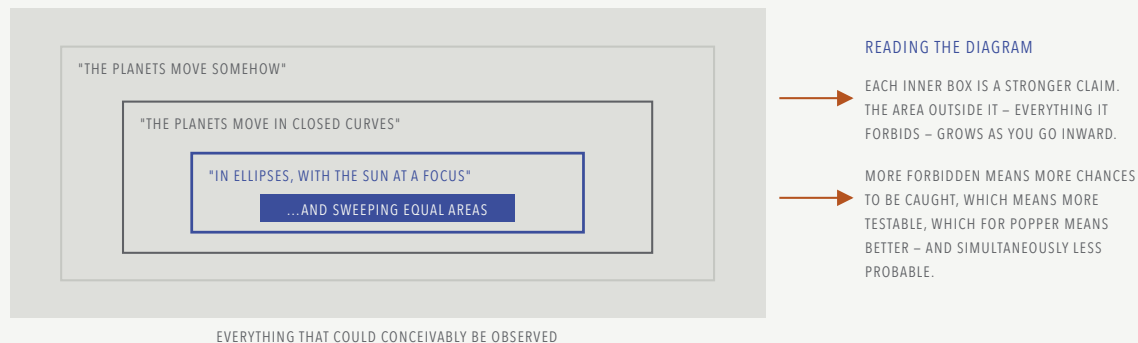
The contrast that produced it is the one from 1919. Popper had been struck that Marx's theory of history, Freud's psychoanalysis and Adler's individual psychology all shared a feature their adherents regarded as their greatest strength: whatever happened, the theory accounted for it. A man pushes a child into a river and a man risks his life to save a child from a river; Freudian theory explains the first as repression and the second as sublimation, and Adlerian theory explains the first as an attempt to prove himself and the second as an attempt to prove himself. Neither could name a piece of human behaviour that would embarrass it. Popper's anecdote about Adler is exact: when he reported a case that seemed not to fit, Adler analysed it in terms of inferiority feelings without having seen the child, and when asked how he could be so certain, said "because of my thousandfold experience" — to which Popper replied that with this new case it was presumably thousand-and-one-fold.

Einstein's theory was in a different position entirely, and the difference was not that it was better supported. It was that Einstein had specified in advance a result that would sink it. General relativity predicted that light passing the sun would be deflected by 1.75 arcseconds — twice the Newtonian figure. Had Eddington's plates shown the Newtonian value, or no deflection, the theory would have been finished, and Einstein said as much. **A theory that takes that risk is doing something a theory that explains everything is not doing.**

PLATE 03 THE TEST POPPER ACTUALLY PROPOSES



The most common misuse of the criterion is to deploy it as a slur, as though "unfalsifiable" meant "rubbish." Popper's own view was the reverse: he thought the metaphysical doctrine of atomism, unfalsifiable for millennia, was one of the most fruitful ideas ever had, and he wrote a defence of Darwinism as a "metaphysical research programme" on precisely these grounds.



SIMPLICITY, ON THIS ACCOUNT, IS NOT AN AESTHETIC PREFERENCE: A SIMPLER LAW IS ONE WITH FEWER ADJUSTABLE PARAMETERS, AND THEREFORE ONE THAT FORBIDS MORE.

This is Popper's answer to the old puzzle of why scientists prefer simple theories. It is not that nature is simple or that simple theories are more likely true – it is that a simpler theory is easier to kill, and being easy to kill is the property that makes a theory worth having.

AND NOTHING UNDERNEATH

If theories are tested against observations, what are the observations tested against? Popper's answer is the most vertiginous thing in his epistemology and the one that most clearly separates him from the empiricists he is often lumped with. Observation statements — he calls them basic statements — are themselves theory-laden, fallible, and accepted by a kind of convention among investigators. They are not certain. They can be, and are, overturned. There is no floor.

"Science does not rest upon solid bedrock. The bold structure of its theories rises, as it were, above a swamp. It is like a building erected on piles... we simply stop when we are satisfied that the piles are firm enough to carry the structure, at least for the time being."

THE LOGIC OF SCIENTIFIC DISCOVERY, §30 – THE IMAGE RECURS THROUGHOUT OBJECTIVE KNOWLEDGE

The consequence is that Popper's system has no foundations anywhere: not in observation, not in reason, not in probability. What holds it up is not a base but a practice — the continual attempt to knock it over. This is why he calls his position *critical rationalism* rather than rationalism, and why he thinks the demand for justification, in epistemology as in politics, is the root error rather than the goal.

Hume's problem: no number of observed instances can justify a universal law, because the inference from "all observed A are B" to "all A are B" is not valid, and any attempt to justify it by experience assumes the very principle in question. Hume accepted the logic and then explained our confidence psychologically — as habit and custom. **Popper accepts Hume's logic and rejects the psychology.** We do not form expectations by repetition; we bring expectations to experience, and experience's only job is to knock them down.

PLATE 05 THE ASYMMETRY THE WHOLE BOOK RESTS ON

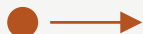
THE CLAIM: ALL SWANS ARE WHITE

CONFIRMING INSTANCES

● ● ● ● ● ● ● ● ● ● ● ● ● ● ● ... AND A MILLION MORE

EVERY WHITE SWAN YOU EVER SEE LEAVES THE CLAIM EXACTLY AS UNCERTAIN AS IT WAS.
LOGICAL VALUE OF A CONFIRMING INSTANCE: NONE. THERE IS NO VALID INFERENCE HERE.

ONE COUNTER-INSTANCE



MODUS TOLLENS
IF T THEN P. NOT-P. THEREFORE NOT-T.

A DEDUCTIVELY VALID INFERENCE – AND THE
ONLY ONE EXPERIENCE EVER GIVES US.

THE CONSEQUENCE

SINCE ONLY REFUTATION IS LOGICALLY AVAILABLE, SCIENCE CANNOT BE A METHOD OF PROVING THINGS. IT CAN ONLY BE A METHOD OF PROPOSING THINGS AND ATTACKING THEM. INDUCTION IS NOT UNRELIABLE – IT IS NOT A THING THAT HAPPENS.

The swan example is Popper's shorthand, not his argument. The argument is that the logical relation between a universal statement and a singular one is asymmetric: universals are not entailed by any number of singulars, but they are contradicted by one. Everything he builds stands on that asymmetry.

THE PROBLEM, SPLIT INTO ANSWERABLE PIECES

Popper's actual method in chapter 1 is to refuse the question as usually posed and break it into separate ones — some of which have a clean answer. The distinction between the third question and the first is the one most readers miss, and it is where the whole dispute with his critics lives.

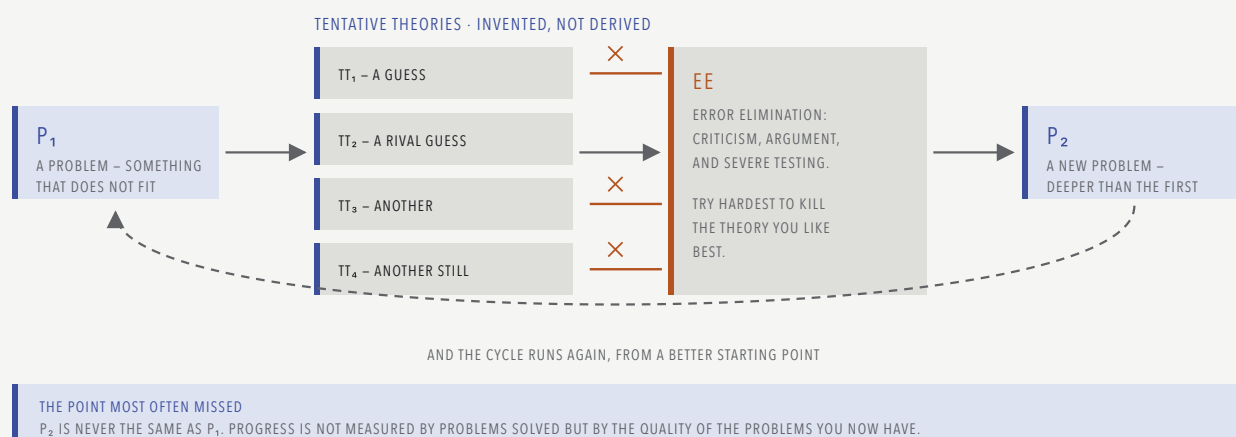
PLATE 06 FOUR QUESTIONS, THREE ANSWERS

QUESTION		POPPER'S ANSWER
L1	CAN A THEORY'S TRUTH BE JUSTIFIED BY EMPIRICAL REASONS?	NO. AND NO PROBABILITY CALCULUS REPAIRS IT.
L2	CAN A THEORY'S FALSITY BE JUSTIFIED BY EMPIRICAL REASONS?	YES. BY MODUS TOLLENS, ASSUMING THE TEST REPORT HOLDS.
L3	CAN A PREFERENCE FOR ONE THEORY OVER ANOTHER BE JUSTIFIED BY EMPIRICAL REASONS?	YES. PREFER THE THEORY THAT HAS SURVIVED THE SEVEREST TESTS SO FAR.
PP	WHICH THEORY SHOULD I RELY ON FOR PRACTICAL ACTION – AND WHY?	THE BEST-TESTED ONE – BECAUSE WE KNOW OF NOTHING BETTER, NOT BECAUSE IT IS PROBABLY TRUE.

Corroboration is a report on past performance, not a forecast. Popper insists it confers no probability of future truth – which is exactly what critics find unsatisfying, since we do in fact build bridges on well-corroborated theories. He would say: yes, and we have no guarantee, and pretending otherwise is what he is objecting to.

If knowledge does not come from experience, where does it come from? Popper's answer is that all knowledge — biological, perceptual, scientific — is produced by the same cycle, and that the cycle starts not with observation but with a **problem**. This four-term schema is the most portable thing in the book, and the piece that later becomes the backbone of David Deutsch's work.

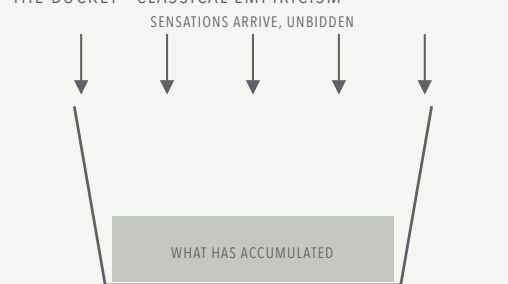
PLATE 07 THE TETRADIC SCHEMA · $P_1 \rightarrow TT \rightarrow EE \rightarrow P_2$



Note where observation enters: nowhere at the start. It appears only inside error elimination, as an instrument for killing theories. This inverts the textbook picture of science – observe, generalise, confirm – and it is the inversion Deutsch later restates as "theory always comes first."

PLATE 08 TWO THEORIES OF THE MIND · THE BUCKET AND THE SEARCHLIGHT

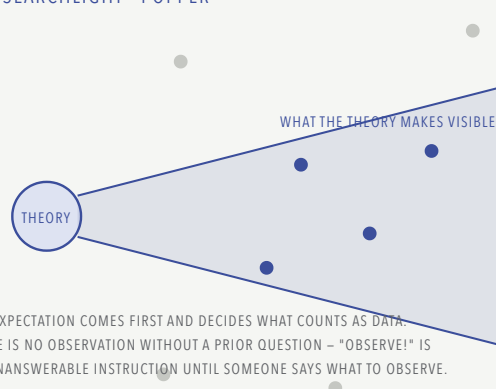
THE BUCKET · CLASSICAL EMPIRICISM



KNOWLEDGE IS WHAT HAS POURED IN AND SETTLED. THE MIND STARTS EMPTY AND IS FILLED BY EXPERIENCE; GENERALISATION HAPPENS BY REPETITION.

POPPER: NO SUCH THING HAS EVER BEEN OBSERVED TO HAPPEN.

THE SEARCHLIGHT · POPPER



THE EXPECTATION COMES FIRST AND DECIDES WHAT COUNTS AS DATA. THERE IS NO OBSERVATION WITHOUT A PRIOR QUESTION – "OBSERVE!" IS AN UNANSWERABLE INSTRUCTION UNTIL SOMEONE SAYS WHAT TO OBSERVE.

The appendix in which this appears is the most readable thing Popper wrote, and it is where his epistemology touches ordinary experience. Every experiment is designed to answer a question; every instrument is built to detect something specific. The grey dots are not hidden – they are simply not being looked for.

The claim the book is named for, and the one philosophers still argue about.

Almost all epistemology since Descartes has asked about the states of a knowing mind: what do *I* believe, and am *I* justified? Popper says this is the wrong object of study. The knowledge that matters — the content of a theory, the logical consequences it has whether or not anyone has noticed them, the problem it leaves unsolved — **is not a mental state and does not depend on there being a mind that holds it.**

PLATE 09 THE THREE WORLDS



WORLD 3 REACHES WORLD 1 ONLY THROUGH A MIND THAT GRASPS IT – AND THEN BUILDS THE AIRCRAFT

WHY THIS IS NOT PLATO

POPPER'S WORLD 3 IS CREATED BY HUMAN BEINGS AND HAS A HISTORY. IT IS NOT ETERNAL, NOT PERFECT, AND NOT DISCOVERED IN A HEAVEN OF FORMS.

The test case Popper offers: if all machines and all subjective knowledge were destroyed but the libraries survived, civilisation could restart. If the libraries were destroyed too, it could not. The difference between those two worlds is exactly what World 3 is.

PLATE 10 INVENTED, AND THEN DISCOVERED



THIS IS THE ARGUMENT FOR WORLD 3'S AUTONOMY, IN ONE EXAMPLE

A PRODUCT OF HUMAN ACTIVITY ACQUIRES CONSEQUENCES ITS MAKERS DID NOT INTEND AND CANNOT REVOKE. THOSE CONSEQUENCES ARE OBJECTIVELY THERE TO BE FOUND, ARGUED ABOUT, AND GOT WRONG. THE SAME HOLDS OF A SCIENTIFIC THEORY: ONCE STATED, IT HAS IMPLICATIONS NOBODY HAS DRAWN, AND THOSE IMPLICATIONS ARE WHAT LATER SCIENTISTS SPEND THEIR CAREERS DISCOVERING.
DEUTSCH'S VERSION: ABSTRACTIONS ARE REAL BECAUSE THEY HAVE EFFECTS AND CANNOT BE REASONED AWAY.

This is the passage that converts most readers, and it is worth sitting with. It gives a precise sense in which mathematics is both invented and discovered, without either Platonism or the claim that mathematics is merely a convention we could have chosen differently.

THE PRACTICAL CONSEQUENCE

If knowledge is a World 3 object, then the right question about a theory is never "how confident is its author?" or "what are their credentials?" but "what does it say, what does it forbid, and where does it break?" Popper thinks the entire tradition of asking after the *sources* of knowledge – reason, sensation, authority, revelation – is a hangover from asking who should rule. The better question is structural: how do we arrange things so that errors get found?

Realism, and What Truth Is For

Theories are guesses about how things actually are — not summaries, not tools.

A recurring worry about Popper is that if we can never justify a theory, and corroboration confers no probability of truth, then talk of truth has quietly dropped out and we are left with an elaborate game. Popper's answer occupies chapters 2, 5 and 9, and it is unambiguous: **theories are attempts to describe a real world that exists independently of us, they are true or false regardless of what anyone believes, and the fact that we cannot certify which is which does not make the distinction idle.**

He positions this between two rivals he thinks are both wrong. The first is *essentialism* — the doctrine that science aims at ultimate explanations, at descriptions of the essences or natures of things that need no further explanation and admit of none. Popper's objection is not that essences are unreal but that the doctrine is a stop sign: it invites us to declare an explanation final, and any explanation declared final is one nobody will try to get behind. Newton's gravitation was treated this way for two centuries; Einstein got behind it.

The second rival is *instrumentalism*, the view — held in various forms by Berkeley, Mach, Duhem and, in Popper's reading, by Bohr — that theories are not descriptions of anything but calculating devices for generating predictions from observations, and so are neither true nor false, only more or less useful. Popper's objection here is sharper and more interesting: an instrument cannot be refuted. A hammer that fails at a task is not shown to be false; it is simply the wrong tool, and you reach for another. If theories are instruments, then the anomalies that drive science forward are not discoveries about the world but merely notes on the limits of applicability, and the entire critical enterprise loses its point.

PLATE 11 THREE VIEWS OF WHAT A THEORY IS		
ESSENTIALISM	INSTRUMENTALISM	CONJECTURAL REALISM
ARISTOTLE, AND NEWTON'S HEIRS	BERKELEY, MACH, DUHEM, BOHR	POPPER
A THEORY IS		
A DESCRIPTION OF THE ESSENCE OF A THING	A CALCULATING DEVICE FOR GENERATING PREDICTIONS	A GENUINE GUESS ABOUT HOW THE WORLD ACTUALLY IS
CAN IT BE REFUTED?		
IN PRINCIPLE NO – AN ESSENCE JUST IS WHAT IT IS	NO. A TOOL THAT FAILS IS NOT FALSE, ONLY UNSUITABLE	YES – AND THAT IS THE WHOLE POINT OF HAVING ONE
SCIENCE AIMS AT		
FINAL EXPLANATIONS THAT NEED NO FURTHER ACCOUNT	SUCCESSFUL PREDICTION AND ECONOMY OF DESCRIPTION	DEEPER EXPLANATIONS, EACH RAISING NEW PROBLEMS
IT GOES WRONG BY		
DECLARING A QUESTION CLOSED	MAKING CRITICISM POINTLESS	NEVER REACHING CERTAINTY
The last row is Popper being fair to his opponents. His position genuinely costs something: it offers no security, no point at which you may stop, and no way to know you have arrived. He regards that as an accurate description of the human situation rather than a defect of the theory.		

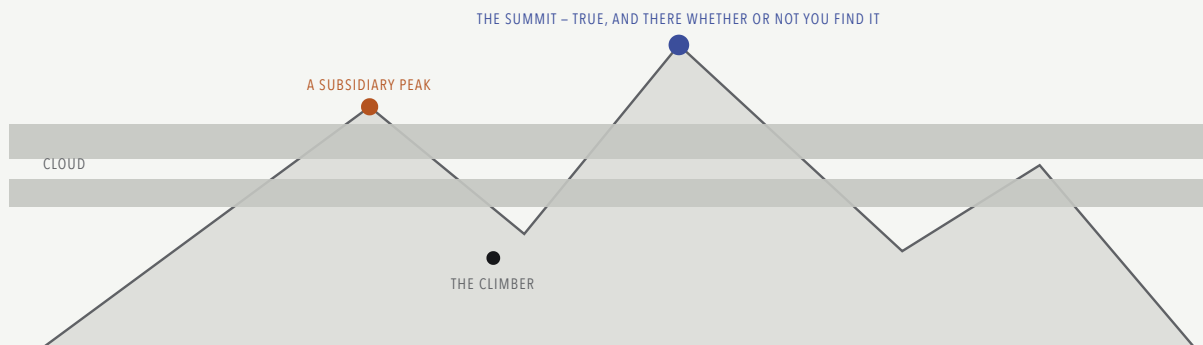
TARSKI, AND WHY POPPER STOPPED BEING EMBARRASSED ABOUT TRUTH

Popper is candid that until the mid-1930s he had been uneasy about the notion of truth, because the correspondence theory — a statement is true if it corresponds to the facts — seemed either empty or paradoxical. What rehabilitated it for him was Alfred Tarski's work of 1933, which showed how to define truth for a formalised language without contradiction, by talking about the object language in a

richer metalanguage. The famous schema — "snow is white" is true if and only if snow is white — looks trivial, and its triviality is the point: it shows that correspondence can be stated precisely and consistently, which is all Popper needed.

What this buys him is a distinction he uses constantly and which the rest of his system would collapse without: the difference between *truth* and *certainty*. Truth is objective and absolute — a statement is true or it is not, and no amount of belief affects this. Certainty is a psychological state, is always attainable and is worth nothing. Popper's whole epistemology is an attempt to keep the first while abandoning the second, and his complaint against most of the tradition is that it conflated them and then, finding certainty unavailable, concluded that truth must be relative too.

PLATE 12 TRUTH AS A REGULATIVE IDEAL – POPPER'S OWN IMAGE



YOU MAY REACH THE TOP AND NOT KNOW IT. YOU MAY BELIEVE YOU ARE AT THE TOP AND BE ON A LESSER PEAK. NEITHER POSSIBILITY PUTS THE SUMMIT IN DOUBT.

Popper uses this figure in chapter 2 and it does a lot of work. It separates the objectivity of truth from our access to it, which is exactly the separation that lets him be a realist about truth and a fallibilist about knowledge at the same time – a combination that looks contradictory until you have the picture.

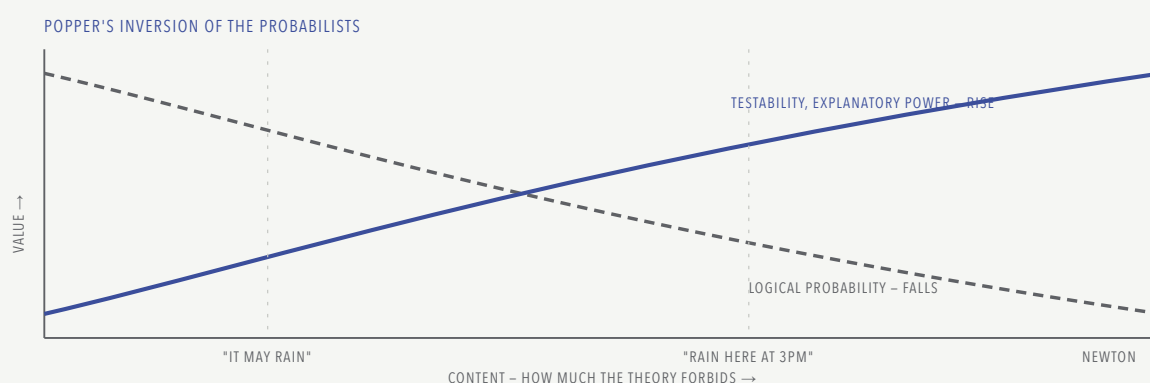
COMMONSENSE REALISM, YES – THE COMMONSENSE THEORY OF KNOWLEDGE, NO

This is the argument of chapter 2, and the title "Two Faces of Common Sense" names the split. Common sense is right that there is a real world of chairs and stars, existing independently of observers, and Popper defends this vigorously against idealism and subjectivism. Common sense is wrong about how we come to know it – it holds the bucket theory, that knowledge is what has flowed in through the senses. Popper's position is that you should keep the first face and discard the second, and that almost all of modern philosophy has done the reverse: grown sophisticated about knowledge while losing its nerve about the world.

Popper's answer is deliberately plain: the aim of science is **to find satisfactory explanations of whatever strikes us as being in need of explanation.**

Prediction is a by-product and a testing device, not the goal. This puts him against the instrumentalists, for whom a theory is a calculating tool that is neither true nor false, and against the positivists, for whom explanation is suspect metaphysics.

PLATE 13 THE COUNTERINTUITIVE RULE: BETTER THEORIES ARE LESS PROBABLE



A TAUTOLOGY FORBIDS NOTHING, IS CERTAINLY TRUE, AND TELLS YOU NOTHING. A LAW OF NATURE FORBIDS ENORMOUSLY MUCH, IS THEREFORE VERY IMPROBABLE, AND IS WORTH HAVING.

This is where Popper and the Carnap school parted permanently. If the aim is high probability, the safest strategy is to say as little as possible – which is the opposite of what good science does. Popper's criterion rewards the theory that sticks its neck out furthest and survives.

PLATE 14 WHAT A GOOD SUCCESSOR THEORY HAS TO DO

1 · ABSORB

EXPLAIN EVERYTHING THE OLD THEORY GOT RIGHT, AND EXPLAIN WHY IT WORKED AS FAR AS IT DID.

2 · CORRECT

SAY SOMETHING DIFFERENT WHERE THE OLD THEORY WAS WRONG, AND SAY WHERE THAT DIFFERENCE SHOWS.

3 · RISK

PREDICT SOMETHING NEW THAT NOBODY WAS LOOKING FOR – THEN SURVIVE THE TEST OF IT.

THE CASE POPPER KEEPS RETURNING TO

NEWTON'S MECHANICS FALLS OUT OF RELATIVITY AS THE LIMITING CASE AT LOW VELOCITY – SO THREE CENTURIES OF SUCCESS ARE EXPLAINED, NOT DENIED.

MERCURY'S PERIHELION HAD BEEN A KNOWN ANOMALY FOR DECADES. RELATIVITY GETS IT EXACTLY RIGHT WITHOUT AN ADJUSTABLE PARAMETER.

LIGHT BENDS NEAR THE SUN – A PREDICTION NOBODY HAD REASON TO MAKE, TESTED IN 1919, AND IT COULD EASILY HAVE FAILED.

Popper's word for the third column is *risk*, and he means it literally: a theory that cannot fail a test is not doing scientific work. The 1919 eclipse is his standing example because Einstein specified in advance what result would sink the theory.

VERISIMILITUDE – THE IDEA, AND THE FORMALISM THAT FAILED

Popper wanted to say that a false theory can still be *closer to the truth* than another false theory, and proposed measuring it as truth content minus falsity content. In 1974 – two years after this book – David Miller and Pavel Tichý independently showed the definition collapses: on it, no false theory can ever be closer to the truth than another. Popper accepted the refutation. The intuition survives and is used constantly; the formal account does not. Read chapter 2's confident passages on verisimilitude with that in mind.

Clouds, Clocks, and the Tree of Knowledge

Where the subtitle — "an evolutionary approach" — earns its place.

Popper's most original move is to treat the growth of knowledge as continuous with the growth of life. Both proceed by variation and selection; both are ways of solving problems set by an environment. The difference is one that matters enormously, and he states it in a sentence that is the emotional centre of the book: **we can let our theories die in our stead.**

PLATE 15 FROM THE AMOEBA TO EINSTEIN – SAME SCHEMA, ONE DIFFERENCE

THE AMOEBA		EINSTEIN
SURVIVE: FIND FOOD, AVOID HARM	PROBLEM	UNDERSTAND: RECONCILE TWO THEORIES THAT CLASH
AN EXPECTATION BUILT INTO ITS STRUCTURE	TENTATIVE THEORY	A HYPOTHESIS STATED IN LANGUAGE, OUTSIDE HIMSELF
THE ENVIRONMENT DOES IT, UNASKED	ERROR ELIMINATION	HE DESIGNS THE TEST THAT COULD DESTROY IT
CANNOT CRITICISE ITS OWN EXPECTATIONS	ATTITUDE TO ERROR	ACTIVELY HUNTS FOR IT – THAT IS THE METHOD
THE ORGANISM DIES WITH IT	IF THE TRIAL FAILS	THE THEORY DIES, AND HE DOES NOT

THIS IS POPPER'S WHOLE CASE FOR CONSCIOUSNESS, LANGUAGE AND CRITICISM HAVING SURVIVAL VALUE: THEY LET SELECTION OPERATE ON OUR GUESSES RATHER THAN ON US.

The continuity claim is strong and the difference is everything. Language makes a theory a World 3 object – something outside the organism that can be examined, attacked and abandoned. Popper thinks this is the single largest change in the history of life on earth.

PLATE 16 CLOUDS AND CLOCKS · AGAINST DETERMINISM AND AGAINST PURE CHANCE



Popper's motive here is moral as much as physical. If determinism is true, then arguing with someone is only pushing their atoms about, and the distinction between persuading and manipulating collapses. He wants an account of the world in which reasons can be causes without being compulsions.

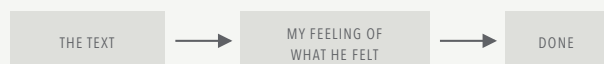
Chapter 4 asks what it means to understand a text, an action, a work of art or a historical event. The traditional answer in the German tradition Popper grew up in — Dilthey's, and later Collingwood's — is empathy: you understand a past agent by re-enacting their mental states, putting yourself in their position, recovering what it felt like to be them. **Popper thinks this is exactly wrong, and that it makes the humanities look unrigorous for no good reason.**

His alternative follows straight from the three worlds. Understanding is not a World 2 operation on someone else's World 2 states, which we have no access to anyway. It is a World 3 operation: you reconstruct the *problem situation* — the problem the author or agent was facing, the background of assumptions they were working within, the options that were available, and what would have counted for them as a solution. Having done that, you can assess whether the move they made was adequate to the problem, which is something you can argue about with evidence and be shown to have got wrong. Empathy offers nothing to argue about.

Popper calls the method *situational analysis* or situational logic, and he offers it as the characteristic method of the social sciences and of history. The explanation of an action is a reconstruction of the situation as the agent saw it, plus the assumption that the agent acted appropriately to it. That assumption — he calls it the rationality principle — is almost certainly false in many individual cases, and Popper says so; he treats it as a deliberately simplifying element, so that when an explanation fails it is the reconstruction of the situation, not human rationality, that we should suspect first.

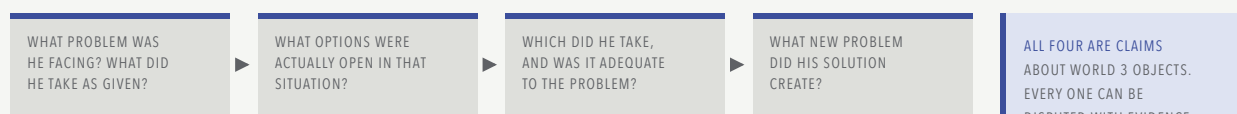
PLATE 17 TWO WAYS TO UNDERSTAND A TEXT OR AN ACTION

RE-ENACTMENT – THE TRADITIONAL ACCOUNT



NOTHING HERE CAN BE CHECKED. IF TWO READERS DISAGREE THERE IS NO PROCEDURE, BECAUSE EACH IS REPORTING A PRIVATE STATE.

PROBLEM-SITUATION RECONSTRUCTION – POPPER



POPPER'S EXAMPLE IS READING A PHILOSOPHER: DO NOT ASK WHAT PLATO FELT, ASK WHAT PROBLEM THE REPUBLIC WAS A PROPOSED SOLUTION TO, WHAT HE WAS ARGUING AGAINST, AND WHETHER THE PROPOSAL SOLVES IT. THEN YOU HAVE SOMETHING A SCHOLAR CAN BE WRONG ABOUT.

This is, incidentally, the method of this document. Summarising a book well means reconstructing what its author took the problem to be – which is why the sections here are organised around Popper's questions rather than around his chapter order.

Why This Is Also a Political Theory

The same mistake, in two disciplines — and the same repair.

Popper never separated his epistemology from his politics, and *Objective Knowledge* keeps gesturing at a connection worked out at length in *The Open Society and Its Enemies*. The link is not an analogy or a metaphor. It is that the two fields have made **the same structural error, which is to ask a question about authority when they should be asking a question about correction.**

In epistemology the bad question is: *what are the sources of our knowledge?* Reason, the senses, intuition, revelation, authority — the tradition has fought over which of these guarantees truth, and Popper's answer is that none does, that knowledge has no pure source, and that the pedigree of an idea is irrelevant to its worth. A theory that arrived in a dream is not thereby worse than one derived from careful observation; both must face the same tests. The right question is not where a belief came from but how we can find out whether it is wrong.

In politics the bad question is: *who should rule?* The best, the wisest, the majority, the workers, the philosopher-kings — Plato posed it in this form and, on Popper's account, poisoned political theory for two thousand years, because every answer to it licenses the concentration of unaccountable power in whoever qualifies. The right question, once you accept that rulers will sometimes be incompetent or malicious, is institutional: *how do we arrange things so that bad rulers can be removed, and can do limited damage while they are there?* On this view democracy is not fundamentally about the wisdom of the people or the legitimacy of majorities. It is defined by a single property — that a government can be got rid of without violence.

PLATE 18 THE SAME REPAIR, APPLIED TWICE

EPISTEMOLOGY	POLITICS
THE QUESTION THAT MISLEADS – BOTH ARE QUESTIONS ABOUT AUTHORITY "WHAT ARE THE SOURCES OF KNOWLEDGE?" EVERY ANSWER CONFERS IMMUNITY FROM CRITICISM	"WHO SHOULD RULE?" EVERY ANSWER CONFERS UNACCOUNTABLE POWER
THE QUESTION THAT WORKS – BOTH ARE QUESTIONS ABOUT CORRECTION "HOW DO WE DETECT AND REMOVE ERROR?" INSTITUTIONS: PEER REVIEW, REPLICATION, OPEN PUBLICATION	"HOW DO WE REMOVE RULERS WITHOUT BLOOD?" INSTITUTIONS: ELECTIONS, COURTS, A FREE PRESS, OPPOSITION

THE PRACTICAL COROLLARY IN BOTH FIELDS IS THE SAME AND POPPER CALLS IT PIECEMEAL: PREFER REFORMS WHOSE CONSEQUENCES YOU CAN OBSERVE AND REVERSE OVER COMPREHENSIVE PLANS WHOSE FAILURES CANNOT BE ISOLATED. A LARGE ENOUGH CHANGE DESTROYS THE VERY FEEDBACK THAT WOULD TELL YOU IT WAS A MISTAKE – WHICH IS THE EPISTEMOLOGICAL CASE AGAINST UTOPIAN POLITICS.

Read this way, *Objective Knowledge* and *The Open Society* are one project. Both hold that we are certain to be wrong, that no authority can be trusted to prevent it, and that the only defence is an arrangement in which being wrong is survivable and discoverable.

PLATE 19 WHERE POPPER SITS AMONG HIS CRITICS

THINKER	UNIT OF APPRAISAL	WHEN AN ANOMALY APPEARS	IS PROGRESS RATIONAL?
POPPER 1902-1994	THE SINGLE THEORY	TAKE IT SERIOUSLY; THE THEORY MAY BE DEAD	YES – IT IS THE MODEL OF RATIONALITY
KUHN 1922-1996	THE PARADIGM	USUALLY TOLERATE IT; PUZZLES ARE THE NORMAL STATE OF AFFAIRS	ONLY WITHIN A PARADIGM
LAKATOS 1922-1974	THE RESEARCH PROGRAMME	PROTECT THE HARD CORE; ADJUST THE BELT AROUND IT	YES, IF THE PROGRAMME IS PROGRESSIVE
FEYERABEND 1924-1994	NOTHING SYSTEMATIC	WHATEVER WORKS; KEEP THE REFUTED THEORY IF IT IS FRUITFUL	"ANYTHING GOES"

LAKATOS AND FEYERABEND WERE BOTH CLOSE TO POPPER AT THE LSE. POPPER'S REPLY TO KUHN IS THAT "NORMAL SCIENCE" DESCRIBES A REAL PHENOMENON AND A REGRETTABLE ONE – THE UNCRITICAL SPECIALIST IS, HE SAYS, A PERSON WHO HAS BEEN BADLY TAUGHT.

The honest position for a reader is that Kuhn describes what scientists do better, and Popper describes what makes the activity worth trusting. They are answering different questions and much of the sixty-year argument between their followers comes from pretending otherwise.

Deutsch is explicit that his epistemology is Popper's, and reading them in order is the single best argument for starting here. Nearly every foundational move in *The Fabric of Reality* and *The Beginning of Infinity* — the rejection of induction and empiricism, fallibilism, knowledge as conjecture plus criticism, problems as the engine — is in this book first. What Deutsch adds is a sharper criterion and a much longer reach.

PLATE 20 THE INHERITANCE, ITEM BY ITEM

POPPER, 1972	DEUTSCH	RELATION
INDUCTION DOES NOT EXIST	IDENTICAL, AND STATED MORE BLUNTLY	TAKEN
KNOWLEDGE GROWS BY CONJECTURE AND CRITICISM	IDENTICAL; THE CORE OF BOTH BOOKS	TAKEN
ALL KNOWLEDGE IS FALLIBLE; CERTAINTY IS UNAVAILABLE	IDENTICAL – "FALLIBILISM"	TAKEN
$P_1 \rightarrow TT \rightarrow EE \rightarrow P_2$	"PROBLEMS ARE INEVITABLE; PROBLEMS ARE SOLUBLE"	TAKEN
THEORIES MUST BE TESTABLE AND RISKY	GOOD EXPLANATIONS ARE HARD TO VARY	SHARPENED
THE AIM OF SCIENCE IS EXPLANATION	EXPLANATION IS THE WHOLE OF IT; REACH IS UNLIMITED	EXTENDED
PROPENSITY INTERPRETATION OF PROBABILITY	EVERETTIAN QUANTUM MECHANICS; MANY WORLDS	DEPARTS

The fifth row is the one worth dwelling on. Popper's criterion is testability: does the theory forbid something observable? Deutsch's is stricter and applies where testing cannot reach – a good explanation has so little slack that changing any detail wrecks it. It rules out the conspiracy theory and the just-so story that Popper's criterion admits when they are carefully stated.

PLATE 21 HARD TO VARY – THE REFINEMENT, IN ONE COMPARISON

EASY TO VARY – A BAD EXPLANATION

"WINTER COMES BECAUSE A GODDESS IS GRIEVING."



SWAP "GODDESS" FOR "GOD", "GRIEVING" FOR "ANGRY", WINTER FOR SUMMER – THE EXPLANATION STILL "WORKS" JUST AS WELL.

EVERY PART IS DETACHABLE, SO THE ACCOUNT EXPLAINS NOTHING: IT WOULD HAVE ACCOMMODATED ANY OUTCOME EQUALLY.

POPPER WOULD CALL THE RIGHT-HAND THEORY MORE TESTABLE. DEUTSCH'S POINT IS THAT TESTABILITY ALONE DOES NOT DISTINGUISH THE TWO WELL ENOUGH: A SUFFICIENTLY CAREFUL MYTH CAN BE MADE TO FORBID SOMETHING. RIGIDITY OF STRUCTURE IS THE PROPERTY THAT ACTUALLY DOES THE WORK.

HARD TO VARY – A GOOD ONE

"WINTER COMES BECAUSE THE AXIS IS TILTED 23.4°."



CHANGE THE TILT AND YOU GET THE WRONG DAY LENGTH. CHANGE THE ORBIT AND THE DATES MOVE. NOTHING CAN BE ADJUSTED IN INSOLATION.

IT ALSO PREDICTS, WITHOUT BEING ASKED TO, THAT THE HEMISPHERES HAVE OPPOSITE SEASONS – WHICH IS HOW YOU WOULD FIND IT WRONG.

Deutsch's example, not Popper's – included because it shows precisely where the tradition moved forward. If you read *The Beginning of Infinity* first and found the "hard to vary" idea striking, this book is where the argument it is repairing was made.

01 You can never falsify a single hypothesis – the Duhem-Quine problem

A test never confronts one theory; it confronts the theory plus assumptions about instruments, background conditions and auxiliary hypotheses. When the result comes out wrong, logic alone does not say which member of the bundle to blame. Popper knew this and answered with methodological rules – do not save a theory by ad hoc rescue – but rules are not deductions, and this is the deepest crack in the edifice.

02 Scientists do not behave this way – Kuhn and Lakatos

Kuhn observed that anomalies are usually tolerated, not treated as refutations, and that theories are abandoned only when a replacement exists. Lakatos tried to rescue Popper by shifting the unit from the single theory to the research programme, with a protected hard core. Popper's reply is that he is describing what makes an activity rational, not what working scientists happen to do.

03 If corroboration says nothing about the future, why act on it?

Popper insists that a well-tested theory is not thereby more likely to be true. But we do board the aircraft. Critics – Salmon most sharply – argue that either corroboration carries some forward-looking weight, in which case something like induction is back, or it does not, in which case Popper has no account of rational action. This is the objection Popper answered least convincingly.

04 The formal theory of verisimilitude is dead

Refuted in 1974 by Miller and Tichý, and Popper conceded. Later repairs exist and none commands agreement. Since "closer to the truth" is doing real load-bearing work whenever Popper explains why we should prefer a better-corroborated theory, this is not a footnote.

05 World 3 is either trivial or mysterious

The complaint from both directions: if World 3 just means "what books and arguments say", it is an unremarkable way of speaking; if it means more, it needs a metaphysics Popper never supplies. He was uninterested in the objection, holding that the autonomy argument – invented sequences producing unbidden consequences – settles it.

06 And a note on the prose

This is a collection of lectures and papers, not a book written straight through, and it repeats itself considerably. Chapters 8 and 9 are technical and can be skipped on a first pass. Popper is also a pugnacious writer who spends real space on grievances against contemporaries.

PLATE 22 THE BOOK ITSELF – WHAT IS IN EACH CHAPTER

CH.	SUBJECT	START HERE?
1	CONJECTURAL KNOWLEDGE – THE SOLUTION TO INDUCTION	ESSENTIAL
2	TWO FACES OF COMMON SENSE – REALISM YES, THE BUCKET NO	ESSENTIAL
3	EPISTEMOLOGY WITHOUT A KNOWING SUBJECT – WORLD 3	ESSENTIAL
4	ON THE THEORY OF THE OBJECTIVE MIND – WORLD 3 APPLIED	SECOND PASS
5	THE AIM OF SCIENCE – EXPLANATION, DEPTH, TESTABILITY	ESSENTIAL
6	OF CLOUDS AND CLOCKS – FREEDOM, DETERMINISM, PLASTIC CONTROL	REWARDING
7	EVOLUTION AND THE TREE OF KNOWLEDGE – THE BIOLOGICAL ANALOGY	REWARDING
8	A REALIST VIEW OF LOGIC, PHYSICS AND HISTORY	TECHNICAL
9	PHILOSOPHICAL COMMENTS ON TARSKI'S THEORY OF TRUTH	TECHNICAL
APP.	THE BUCKET AND THE SEARCHLIGHT – THE CLEAREST THING HERE	READ FIRST

A workable route for a first reading: the appendix, then chapter 1, then 3, then 5, then 7. That is roughly half the book and contains all of the argument that matters downstream.

01 **There is no such thing as induction, and nothing is lost.**

We
never
reason
from
instances
to laws.
We
guess
laws
and
then
hunt for
the
instance
that
kills
them.
The
asymmetry
between
confirmation
and
refutation
is doing
all the
logical
work.

02 **Observation is always selective.**

"Observe!"
is not
an
instruction
anyone
can
follow. A
theory,
an
interest
or a
problem
must
come
first to
say
what
counts
as data
– the
searchlight,
not the
bucket.

03 **Ask what a theory forbids, not who holds it.**

The
question
about
sources
–
reason,
senses,
authority
– is a
leftover
from
asking
who
should
rule.
Replace
it with
an
institutional
question:
how do
we
arrange
things
so
errors
are
found
fast?

04 **The best theories are the least probable.**

Content,
testability
and
explanatory
depth
all rise
together,
and
logical
probability
falls as
they do.
Safety
and
informativeness
pull in
opposite
directions.

05 Progress is measured in better problems.

P_2 is
never
 P_1 -A
field
that has
stopped
generating
new
and
harder
problems
is not
consolidating;
it has
stopped.

06 Knowledge exists outside heads, and that is what makes criticism possible.

Once a
guess is
written
down it
becomes
an
object
others
can
attack.
This, for
Popper,
is what
language
is for
and
why it
changed
everything.

The
one-
sentence
summary
of the
whole
book,
and the
reason
Popper
thought
his
epistemology
was also
a moral
and
political
position.

COLOPHON

Objective Knowledge · Karl Popper

A reader's summary of *Objective Knowledge: An Evolutionary Approach*, Clarendon Press 1972, revised 1979. Every plate was drawn for this document. Where later work has undercut a position – most importantly the formal theory of verisimilitude – it is marked in the text rather than left standing. The Deutsch comparisons draw on *The Fabric of Reality* and *The Beginning of Infinity*.

THE ARGUMENT IN ONE LINE

We do not derive knowledge from experience.
We invent it, and experience kills the bad guesses.

PLATES

13 hand-drawn figures
All vector; all legible at print size

READ NEXT

Conjectures and Refutations · Popper
The Beginning of Infinity · Deutsch

ALSO IN THIS SERIES

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